CPEN 416

# Lecture 09: Grover's algorithm

Tuesday 13 October 2026

# Announcements

34 drop p2 → final denom: 25.5

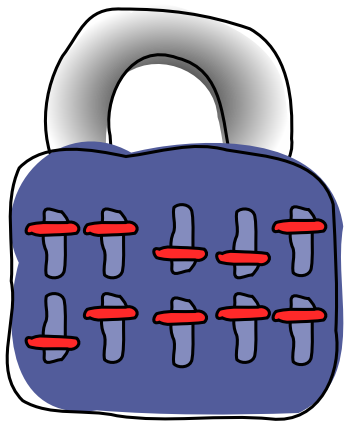
- Midterm available for pick up at my office
- Practice assignment 2 available
- Project details available later this week

Vazirani

# Last time

We modeled breaking a (binary) lock as a function:

$$f(\vec{x}) = \begin{cases} 1 & \vec{x} = \vec{s} \\ 0 & \text{otherwise} \end{cases}$$



← correct combo $\vec{s}$

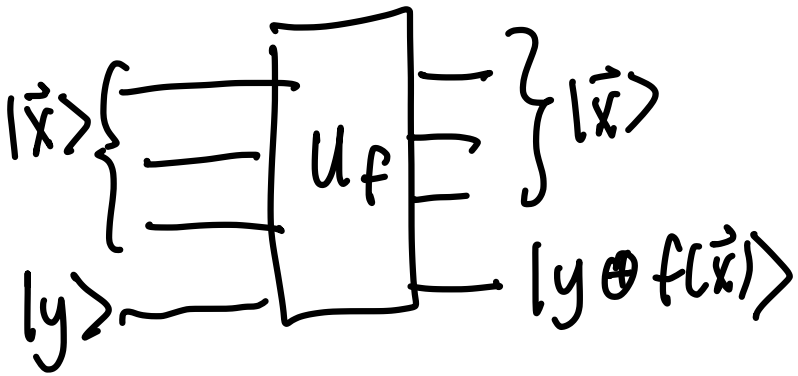
Trying combination $\mathbf{x}$ modeled as querying an *oracle* that evaluates $f(\mathbf{x})$. The number of queries is an algorithm's *query complexity*.

Image credit: Codebook node A.1

# Last time

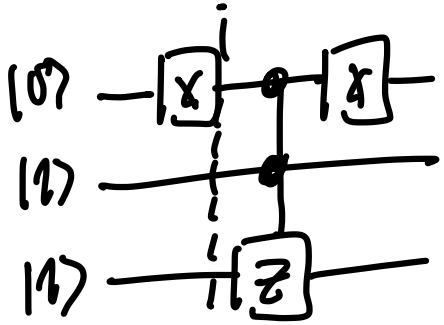
We expressed oracle queries as quantum circuits in two ways.

v1: $U_f|\vec{x}\rangle|y\rangle = |\vec{x}\rangle|y \oplus f(\vec{x})\rangle$



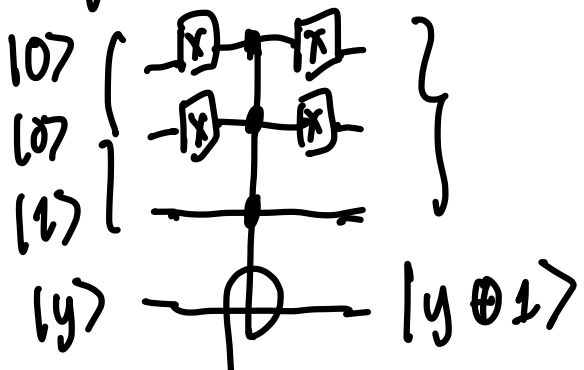

v2: $U_f|\vec{x}\rangle = (-1)^{f(\vec{x})}|\vec{x}\rangle$

convert to $|111\rangle$




$\rightarrow \vec{s} = ??$ 1011

Quiz: given $\vec{s} = 001$



# Last time

We saw how phase kickback can be used to obtain information about multiple evaluations of $f(x)$ with a single query.

(oracle v1)

$$|x\rangle|-\rangle = \frac{1}{\sqrt{2}}|x\rangle|0\rangle - \frac{1}{\sqrt{2}}|x\rangle|1\rangle$$

$$U_f(|x\rangle|-\rangle) = \frac{1}{\sqrt{2}}U_f(|x\rangle|0\rangle) - \frac{1}{\sqrt{2}}U_f(|x\rangle|1\rangle)$$

$$= \frac{1}{\sqrt{2}}|x\rangle|0\oplus f(x)\rangle - \frac{1}{\sqrt{2}}|x\rangle|1\oplus f(x)\rangle$$

if $f(x)=0$: $|x\rangle|-\rangle \longmapsto |x\rangle|-\rangle$

if $f(x)=1$: $|x\rangle|-\rangle \longmapsto -|x\rangle|-\rangle$

$$\Big\} U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$$

$$U_f|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)|-\rangle \longrightarrow U_f|+\rangle|-\rangle = (-1)^{f(0)}\frac{1}{\sqrt{2}}|0\rangle|-\rangle + (-1)^{f(1)}\frac{1}{\sqrt{2}}|1\rangle|-\rangle$$

# Last time

We saw how phase kickback can be used to obtain information about multiple evaluations of $f(x)$ with a single query.

$$U_f(|+\rangle|-\rangle) = (-1)^{f(0)}\left[\frac{1}{\sqrt{2}}|0\rangle|-\rangle + \frac{1}{\sqrt{2}}(-1)^{f(0)\oplus f(1)}|1\rangle|-\rangle\right]$$

constant func: $f(0)\oplus f(1) = 0$

$$\Rightarrow U_f|+\rangle|-\rangle = |+\rangle|-\rangle$$

balanced func: $f(0)\oplus f(1) = 1$

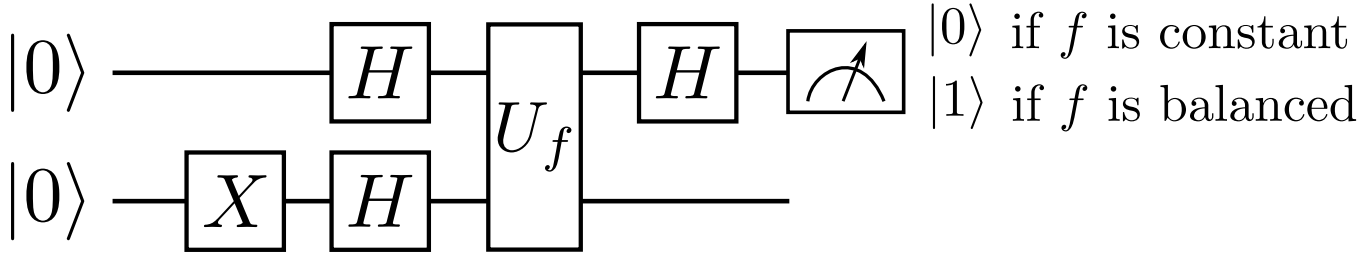
$$U_f|+\rangle|-\rangle = |-\rangle|-\rangle$$

# Last time

We applied Deutsch's quantum algorithm to determine if a function is *constant* or *balanced* using one oracle query

| Name | Action | Name | Action |
|---|---|---|---|
| $f_1$ | $f_1(0) = 0$<br>$f_1(1) = 0$ | $f_2$ | $f_2(0) = 1$<br>$f_2(1) = 1$ |
| $f_3$ | $f_3(0) = 0$<br>$f_3(1) = 1$ | $f_4$ | $f_4(0) = 1$<br>$f_4(1) = 0$ |

constant
$f(0) \oplus f(1) = 0$
balanced



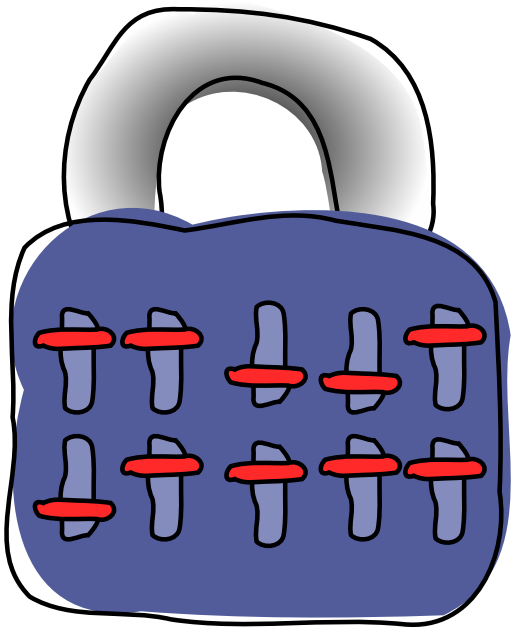

This is a quantum speedup; classical case requires 2 queries.

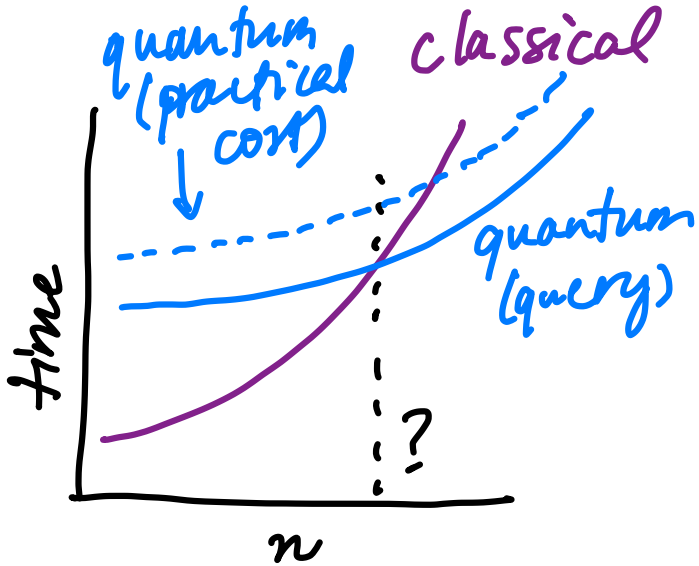
# Learning outcomes

- Describe the strategy of amplitude amplification
- Derive the query complexity of Grover's search algorithm
- Implement Grover's search algorithm in PennyLane

# Grover's quantum search algorithm

Let's break that lock!





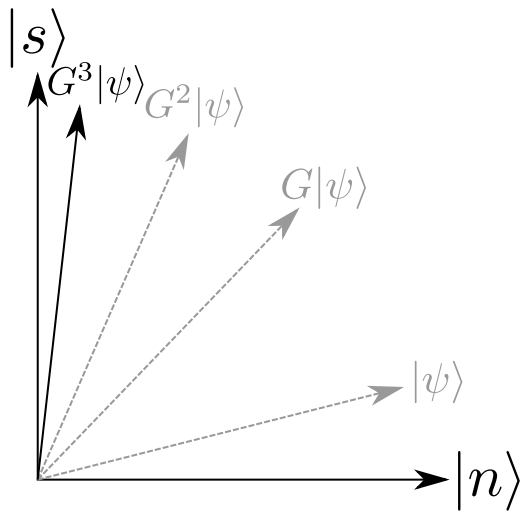

Classical: in the worst case, $2^n$ oracle queries

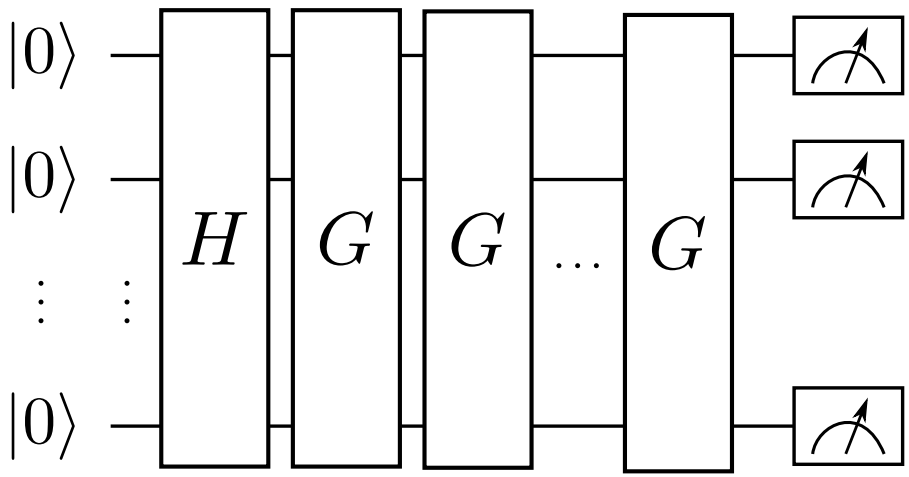
Quantum: $O(\sqrt{2^n})$ queries with Grover's algorithm

Image credit: Codebook node A.1

# Grover's quantum search algorithm

Grover's search algorithm starts with a uniform superposition, then *amplifies* the amplitude of the state corresponding to the solution.



# Grover's quantum search algorithm

In other words, we want to go from the uniform superposition

$|\psi\rangle = (H \otimes H \otimes \cdots \otimes H)|0\cdots 0\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2}^n} \sum_{\vec{x} \in \{0,1\}^n} |\vec{x}\rangle$

$n$ Hadamards on $n$ qubits

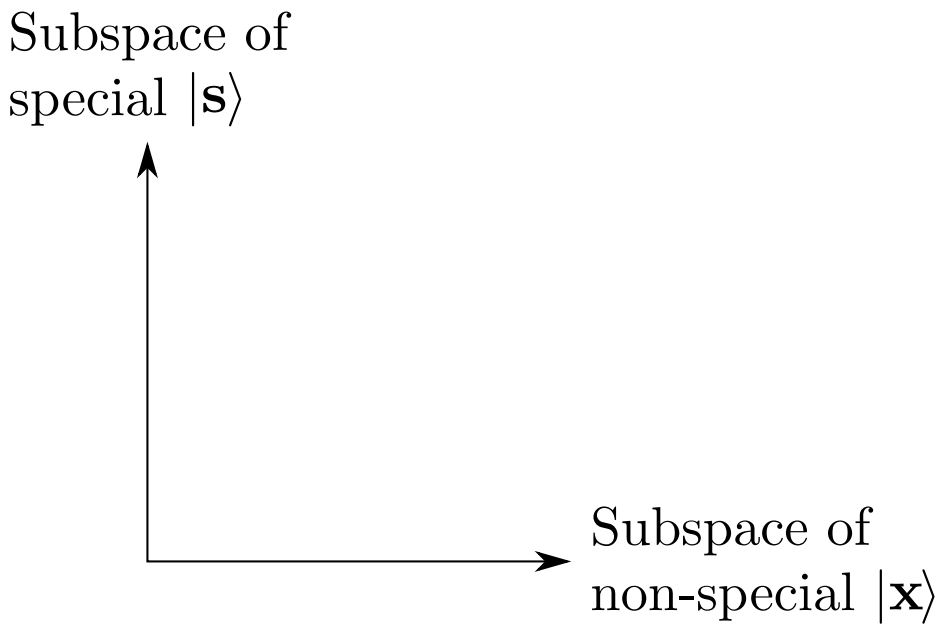
to something like this:

(big amplitude) $\cdot\, |\vec{s}\rangle$

$+$ (small amplitude) $\sum_{\vec{x} \neq \vec{s}} |\vec{x}\rangle$

# Amplitude amplification: geometric visualization

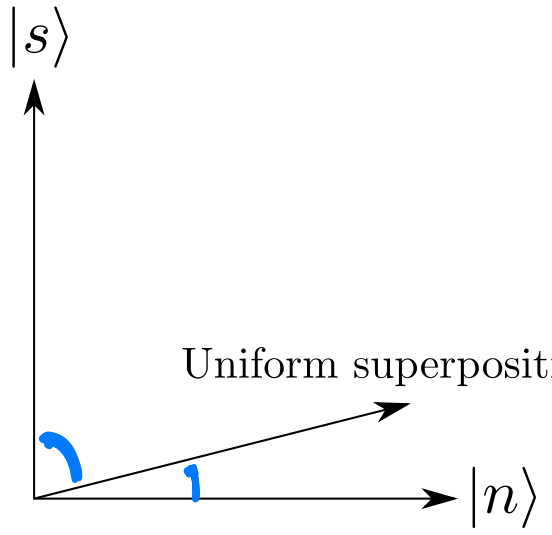
$$\langle \vec{s} | \vec{x} \neq \vec{s} \rangle = 0$$




Partition the computational basis states into two subspaces:

1. The special state $|\mathbf{s}\rangle$
2. All the other states

# Amplitude amplification: geometric visualization




$|s\rangle = |\vec{s}\rangle$

$|n\rangle = \frac{1}{\sqrt{2^n - 1}} \sum_{\vec{x} \neq \vec{s}} |\vec{x}\rangle$

strong overlap

We can write the uniform superposition in terms of $|s\rangle$ and $|n\rangle$:

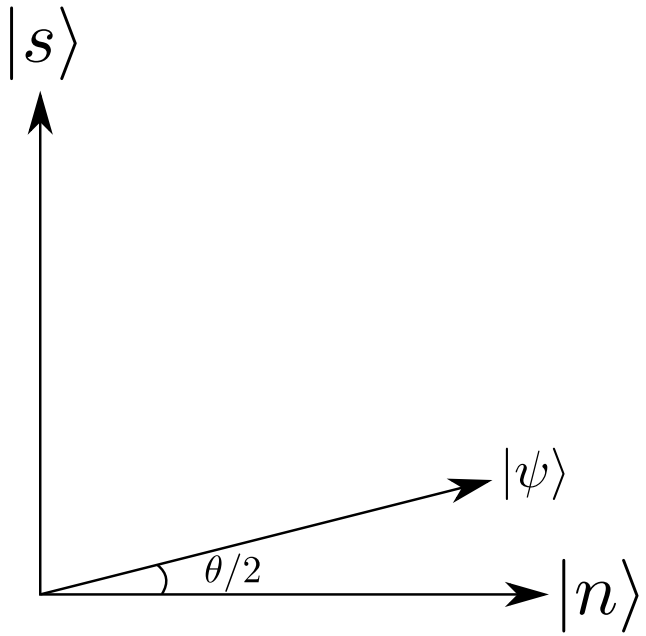
$$|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{x} \in \{0,1\}^n} |\vec{x}\rangle = \frac{1}{\sqrt{2^n}} |\vec{s}\rangle + \frac{1}{\sqrt{2^n}} \sum_{x \neq s} |\vec{x}\rangle$$

$$= \frac{\sqrt{2^n - 1}}{\sqrt{2^n}} \cdot |n\rangle + \frac{1}{\sqrt{2^n}} |s\rangle$$

$$|\langle s|\psi\rangle|^2 = \frac{1}{2^n}$$

$$|\langle n|\psi\rangle|^2 = \frac{2^n - 1}{2^n}$$

# Amplitude amplification: geometric visualization



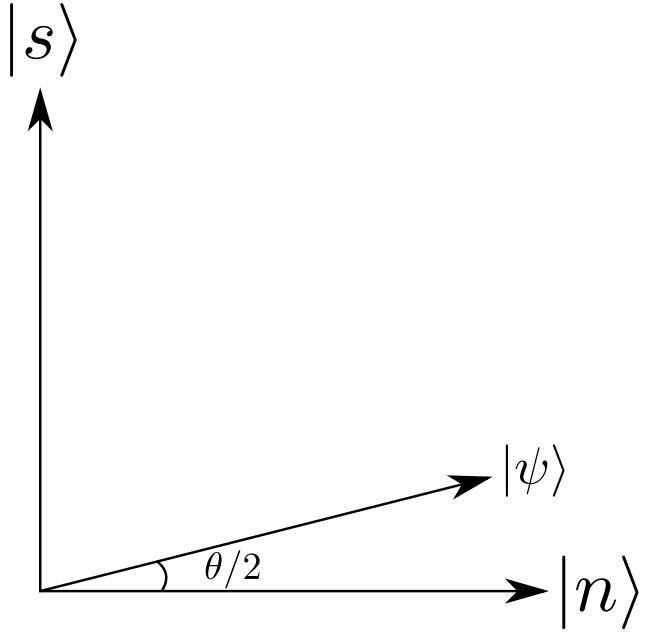

Instead of working with these complicated coefficients

$$|\psi\rangle = \frac{1}{\sqrt{2^n}}|s\rangle + \frac{\sqrt{2^n-1}}{\sqrt{2^n}}|n\rangle,$$

rexpress in terms of an angle $\theta$:

$$= \sin\frac{\theta}{2}|s\rangle + \cos\frac{\theta}{2}|n\rangle$$

# Amplitude amplification: geometric visualization



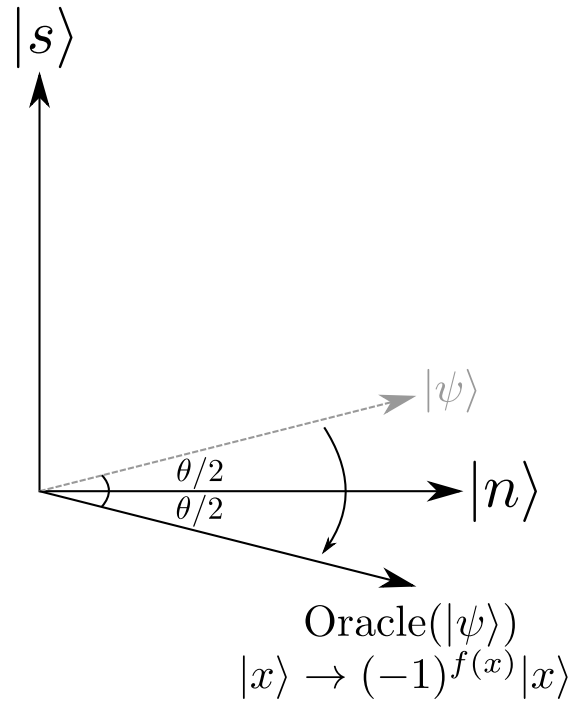

Apply operations to increase the amplitude of $|s\rangle$ while decreasing that of $|n\rangle$.

Two steps:

1. Apply the oracle $O$ to 'pick out' the solution
2. Apply a 'diffusion operator' $D$ to adjust the amplitudes.

$$O|\vec{x}\rangle = (-1)^{f(\vec{x})}|\vec{x}\rangle$$

# Amplitude amplification: geometric visualization



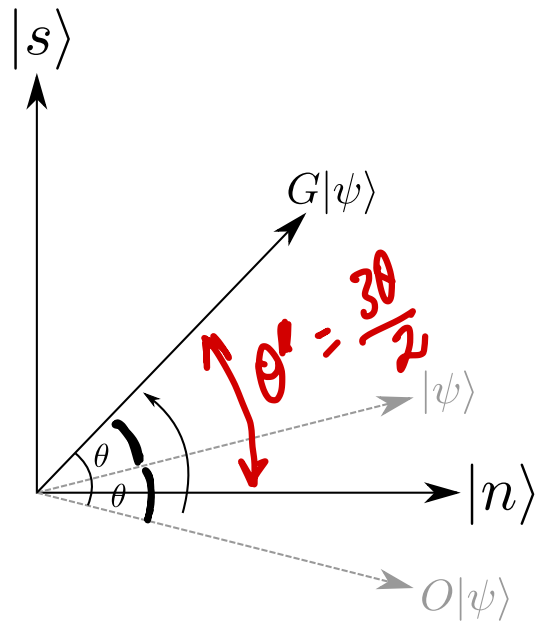

The oracle *flips* the amplitude of the special basis states in $|\psi\rangle$.

We can visualize this as a *reflection about the subspace* of non-special elements.

Exercise: how does this affect the uniform superposition state?

$|\psi\rangle = \sin\frac{\theta}{2}|s\rangle + \cos\frac{\theta}{2}|n\rangle \longmapsto -\sin\frac{\theta}{2}|s\rangle + \cos\frac{\theta}{2}|n\rangle$

# Amplitude amplification: geometric visualization

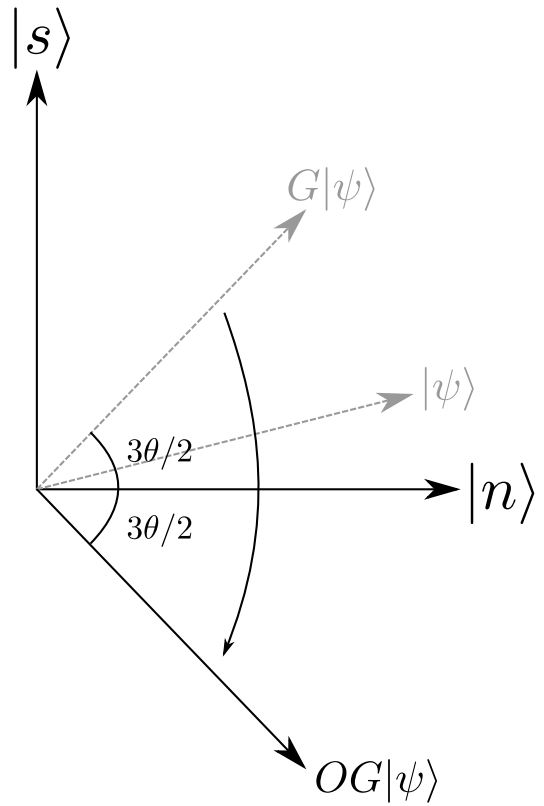



The diffusion operator is a bit less intuitive - it performs a *reflection about the uniform superposition* state.

Combined action $G = DO$ sends

$$G\left(\sin\left(\frac{\theta}{2}\right)|s\rangle + \cos\left(\frac{\theta}{2}\right)|n\rangle\right) = \sin\left(\frac{3\theta}{2}\right)|s\rangle + \cos\left(\frac{3\theta}{2}\right)|n\rangle$$

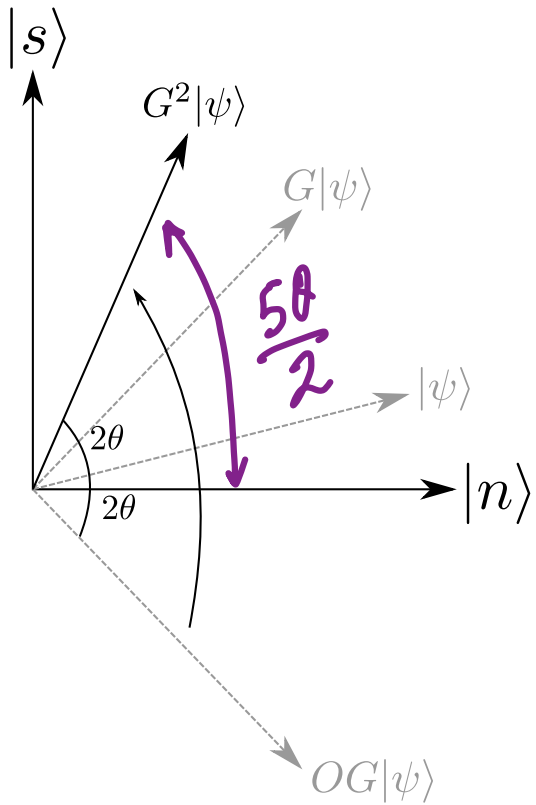
# Grover's algorithm: geometric visualization




Grover's algorithm works by repeated application of $G = DO$ (called the Grover iteration).

Apply the oracle and reflect about the non-special elements.

# Grover's algorithm: geometric visualization

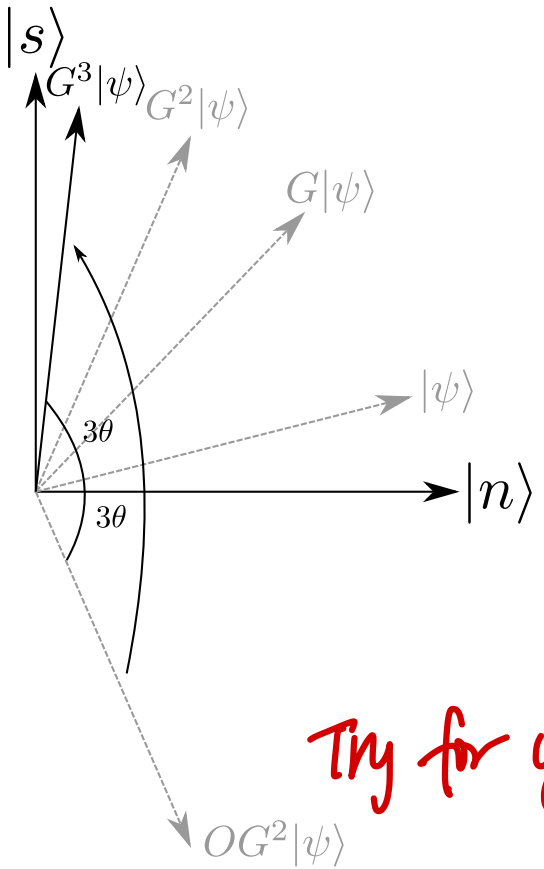



Apply the diffusion operator and reflect about the uniform superposition to boost the amplitude of the special state.

# Grover's algorithm: geometric visualization

After $k$ Grover iterations we will have the state

$$G^k|\psi\rangle = \sin\left(\frac{(2k+1)\theta}{2}\right)|s\rangle + \cos\left(\frac{(2k+1)\theta}{2}\right)|n\rangle$$




**Exercise**: what $k$ *maximizes* the probability of measuring $|s\rangle$?

compute prob, take derivative, and solve for $k$:

$$k = \left\lfloor \frac{\pi}{4}\sqrt{2^n} \right\rfloor$$

Try for yourself: multi-solution case (m solutions)

write $s = \frac{1}{\sqrt{2^m}} \sum_{\text{solutions}} |s\rangle$

# Next time

Content:

- Introduction to quantum compilation and resource estimation

Action items:

1. Start picking project group
2. Practice assignment 2

Recommended reading:

- For this class: Codebook module GA; Nielsen & Chuang 6.1
- For next class: Nielsen & Chuang 4.5